SANATH KUMAR

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PROFESSIONAL SUMMARY

Junior Software Developer specializing in Python, Django Rest Framework, and React.js. Hands- on experience in developing full-stack web applications. Expert in building secure REST APIs that improved backend response times by 60% and optimizing MySQL databases to reduce page load speeds by 40%. Proficient in Agile workflows, Git version control, and Docker containerization.

SKILLS

Languages & Frameworks: Python, Django (DRF), React.js, Redux, JavaScript, HTML, CSS, Bootstrap, Tailwind CSS
Database: MySQL, MongoDB
Tools & Platforms: Visual Studio Code, PyCharm**,** Git, GitHub, Docker, Kubernetes, Linux
Core Competencies: REST APIs, Authentication & Authorization, Database Optimization, Cloud Deployment, Agile, CI/CD (basic)

ACADEMIC PROJECT

Privacy-Critical Cloud Storage Framework – Python, Encryption

- Engineered a robust searchable encryption framework to facilitate high-security data storage in cloud environments.
- Implemented complex dual hash indexing and incidence matrices, enabling high-speed search queries directly on encrypted data.
- Developed an end-to-end encryption pipeline for secure file distribution, successfully mitigating risks of unauthorized data access.
- Validated system performance through rigorous scalability and security testing under high-load conditions.

PERSONAL PROJECTS

Pulse Guard – Real-Time Uptime Monitoring Platform

- Buit a full-stack monitoring platform using React, Express, Node.js, MongoDB, and supports health services across 7 protocols of HTTP/S, TCP, UDP, DNS, PING, SMTP, and SSL with live monitor dashboards, and incident tracking.
- Developed a real-time event pipeline using Redis Streams + Socket.IO for instant status updates with guaranteed delivery, replacing Pub/Sub to prevent event loss under load.
- Designed a scalable job scheduling via BullMQ and Redis with master-worker coordination, automatic retries, and graceful shutdown orchestrating with parallel health checks across all monitors.
- Implemented a security-hardened API with JWT-authenticated Web Sockets, Helmet CSP, rate limiting, HTTP parameter pollution protection, and role-based access controls.
- Enabled a smart alerting engine that evaluates health states, computes uptime/MTTD metrics, and dispatches email alerts via Brevo SMTP.

AI Video Transcriber Tool – Python, Whisper, MoviePy

- Engineered a robust pipeline to automate audio extraction from high-definition video formats (.mp4, .mkv, .mov) using MoviePy, ensuring zero loss in audio fidelity before transcription.
- Used OpenAI's Whisper AI to ensure high accuracy and added batch-processing to handle multiple files at once.
- Created a feature to export transcriptions into TXT and JSON formats for easy sharing.

E-Commerce Platform – Python, Django, DRF, MongoDB

- Developed a full-stack e-commerce application focused on seamless cross-platform performance and custom UI components.
- Implemented secure checkout flows by seamlessly integrating the PayPal API and Cash on Delivery functionality.
- Integrating Brevo SMTP for instant password resets and contact form notifications.
- Optimized backend infrastructure by building high-performance REST APIs, resulting in a 60% reduction in response time.
- Developed a scalable inventory management portal that consolidated disparate data sources, reducing manual data entry time by 80% and improving real-time stock tracking accuracy.
- Refined complex MongoDB database queries, which successfully decreased front-end page load times by 40%.

EDUCATIONAL QUALIFICATIONS

Sree Chaitanya College of Engineering, Karimnagar **2017 - 2021**

M.C.A in Computer Science

Sree Chaitanya Degree College, Karimnagar **2014 - 2017**

B.SC in Computer Science

SR Junior College, Karimnagar **2012 – 2014**

Intermediate

Bharateeya Vidhya Kendra High School, Karimnagar **2011 - 2012**

High School

CERTIFICATES

- Python 3 Course: Beginner to Advanced |Udemy
- HTML, JavaScript, & Bootstrap |Udemy
- MySQL for Absolute beginners |Udemy

STRENGTHS

- Positive attitude and strong work ethic.
- Ability to learn quickly and adapt to new technologies.
- Strong attention to detail and ability to multitask.